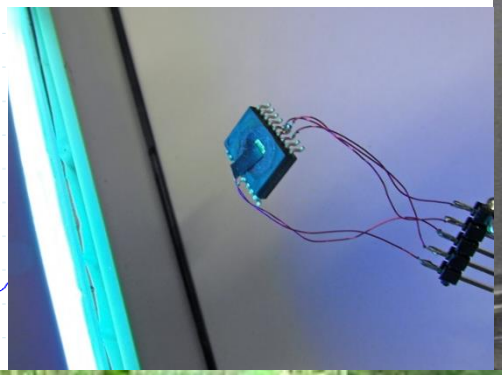
Fault injection 2

David Oswald

Code:



Goals of this lecture

After this lecture, you will …

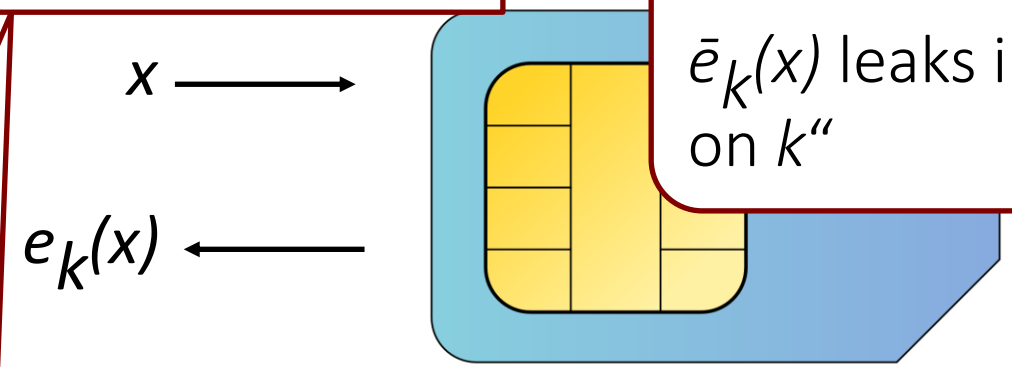
- know physical methods to inject faults from software (and some other ways)
- understand techniques to break generic code with fault injection, not just crypto

Embedded reality:

Adversa
controls

2. Control: fault injection

- Power glitch
- Clock glitch
- (Laser) light
- …



Intuition: "Faulty output $\bar{e}_k(x)$ leaks information on k"

Shortcomings:

- Physical access
- Crypto-specific

A short history of fault injection

1997: “On the importance of checking cryptographic protocols for faults” (Boneh et al.)

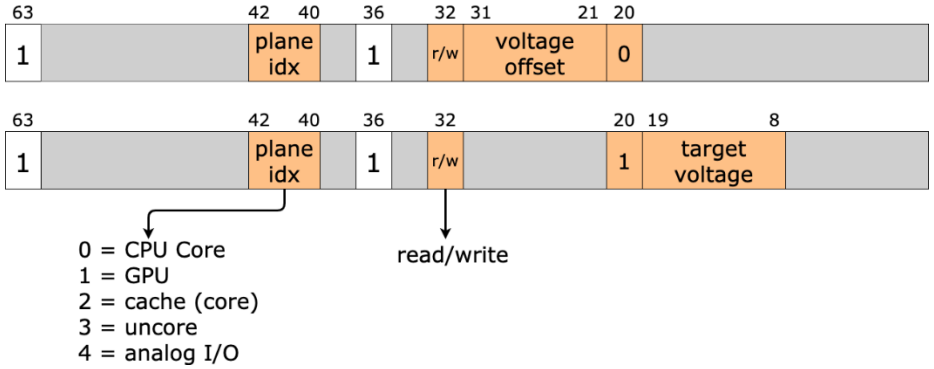
… various hardware fault attacks …

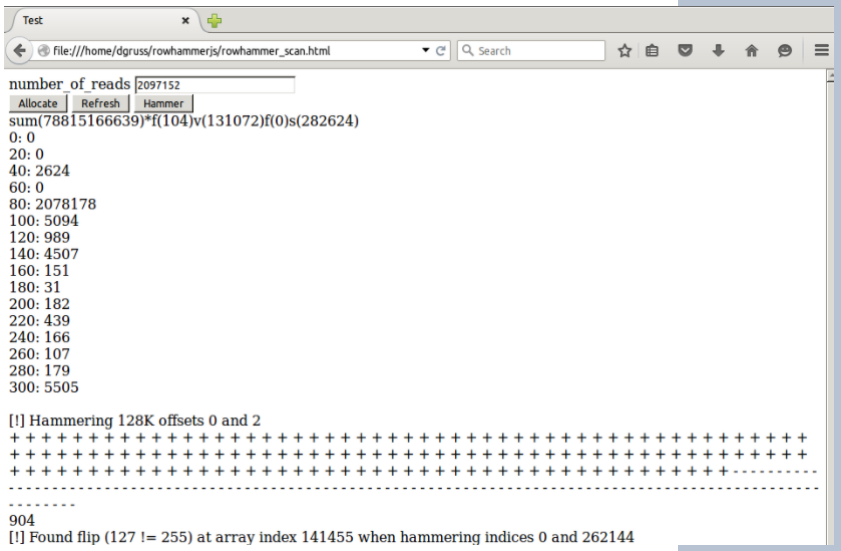
2014: “Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors” (Kim et al.)

2015: “Exploiting the DRAM Rowhammer bug to gain kernel privileges” (Seaborn et al.)

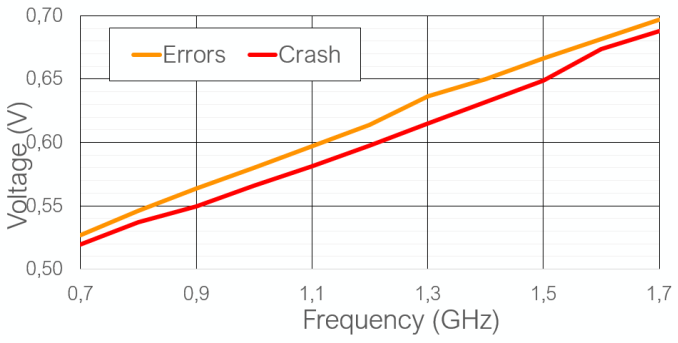
2017: “CLKSCREW: Exposing the Perils of Security Oblivious Energy Management” (Tang et al.)

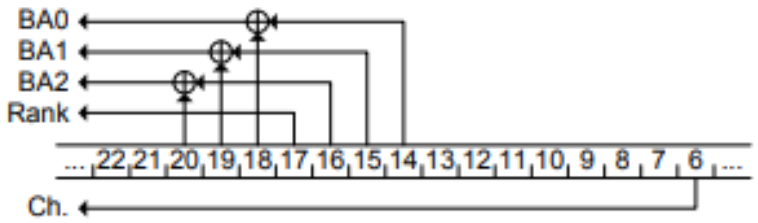
… many more software fault attacks …





Software-based fault injection

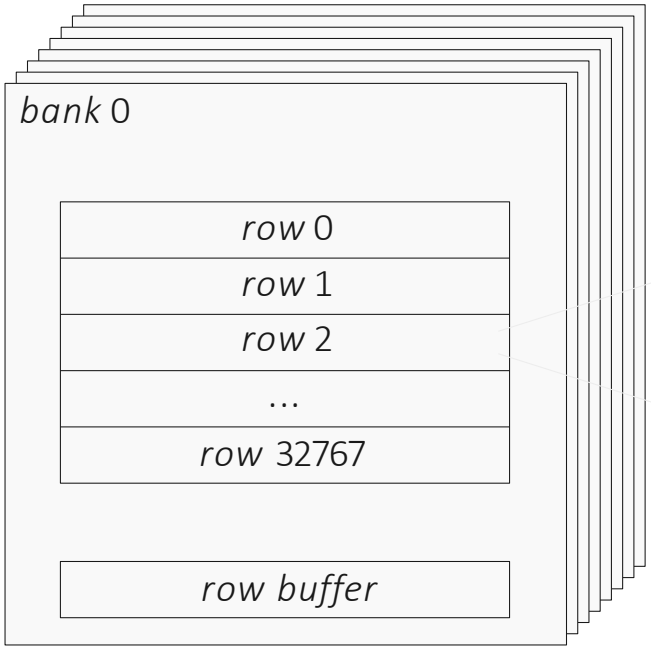




Software-based fault injection

- Physical attackers are a realistic threat for smartcards, but not for general-purpose PCs
- Assume an attacker has “just” code execution: what can we do?
- Idea: “magic” instruction sequences (that can be run in user mode): The **Rowhammer** issue

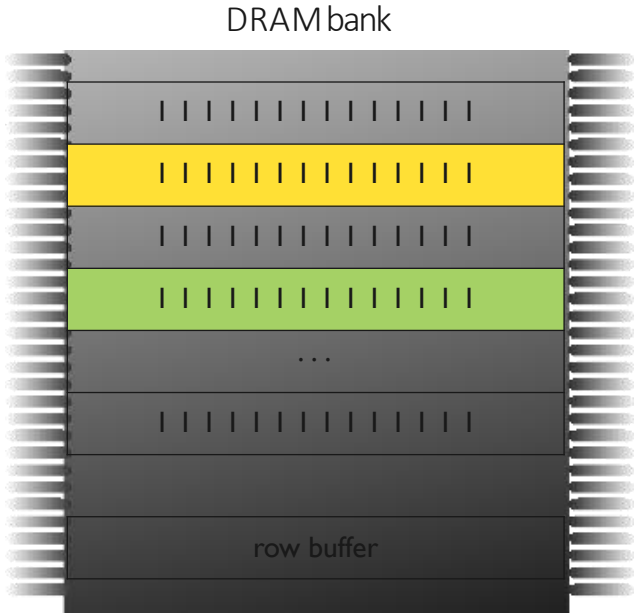
Rowhammer: DRAM organisation



64k cells
1 capacitor,
1 transistor each

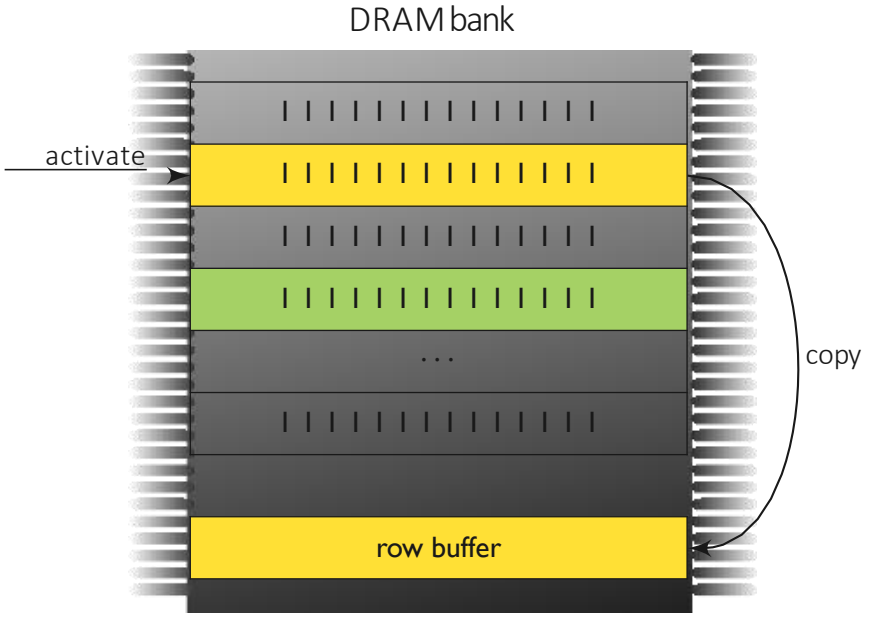
Slides: Thanks to Daniel Gruss

The Rowhammer effect



- Cells leak → need refresh
- Max. refresh interval to guarantee data integrity
- Cells leak faster upon proximate accesses → Rowhammer

The Rowhammer effect



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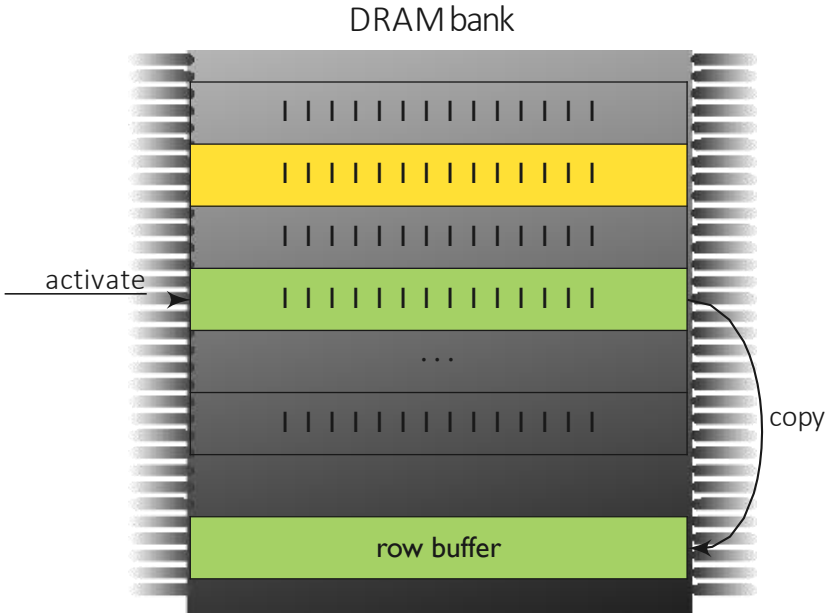

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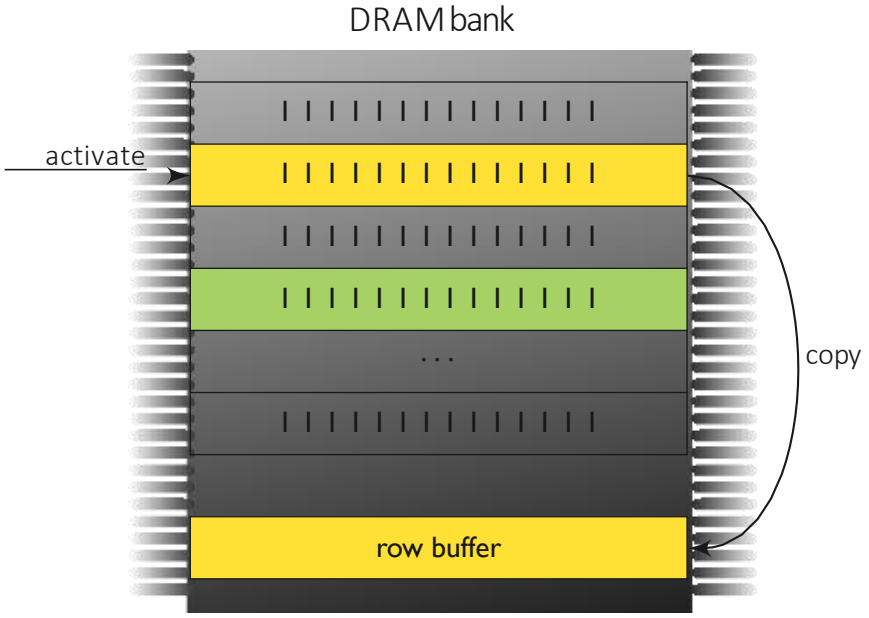

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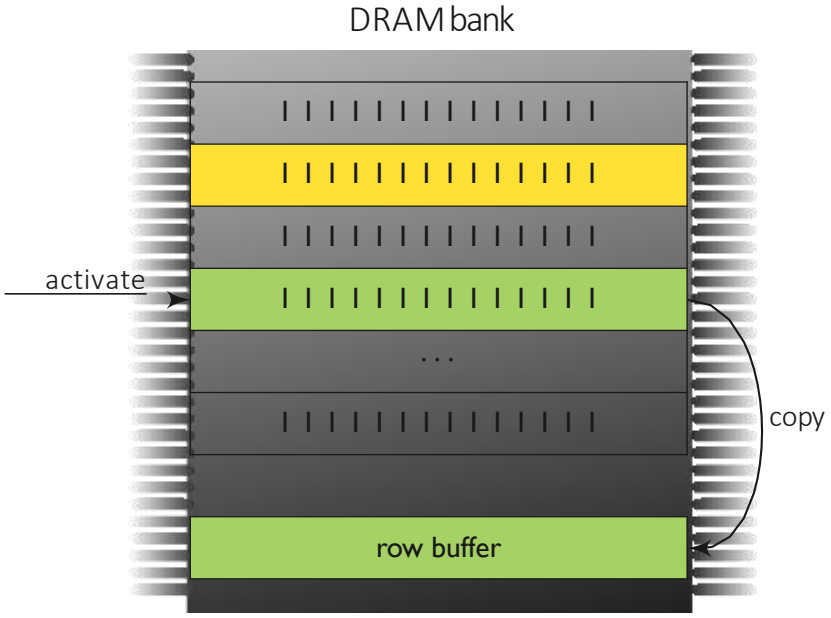

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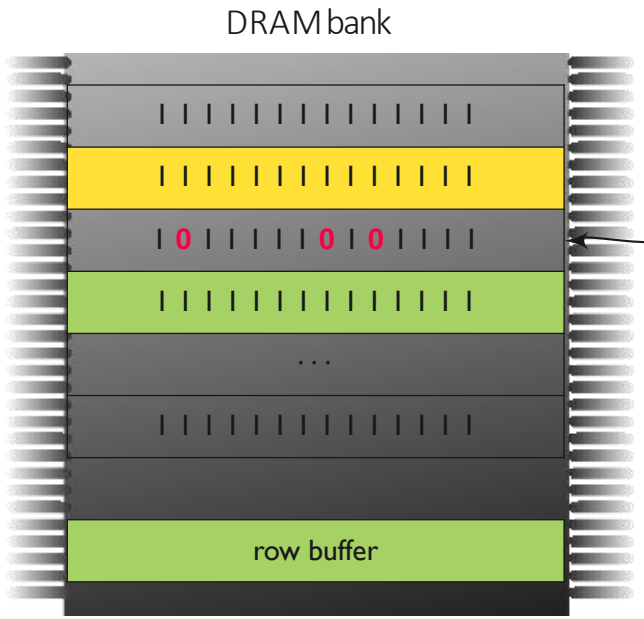
The Rowhammer effect



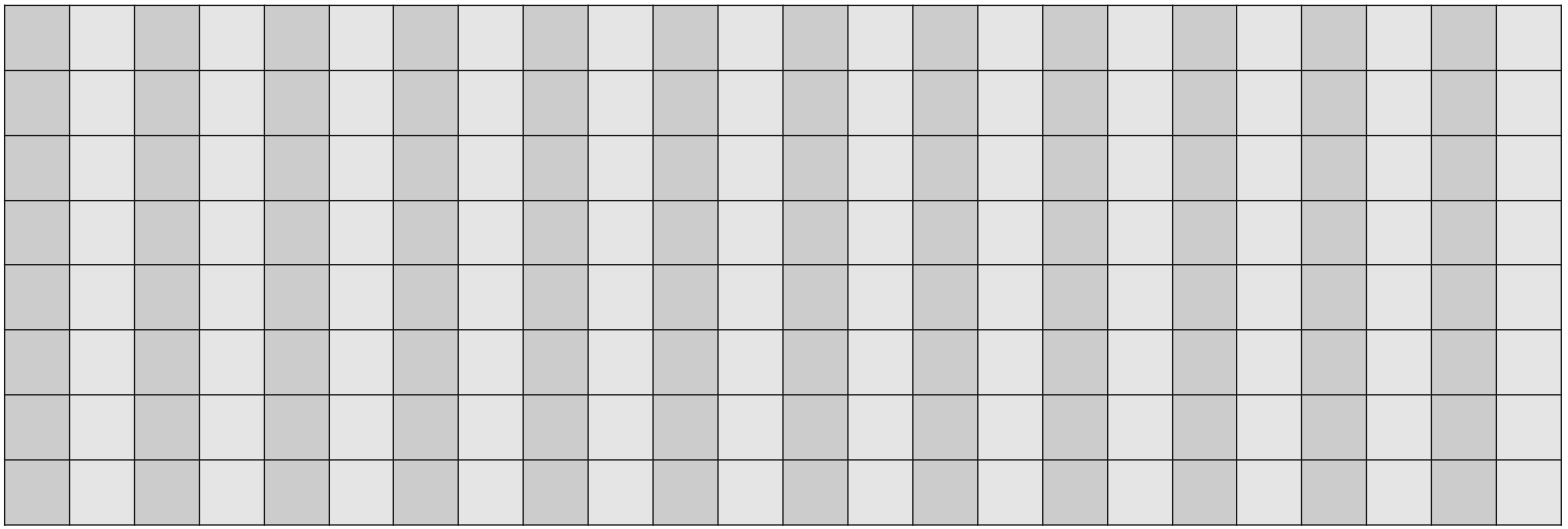
- Cells leak → need refresh
- Max. refresh interval to guarantee data integrity
- Cells leak faster upon proximate accesses → Rowhammer

Rowhammer prerequisites

We need:

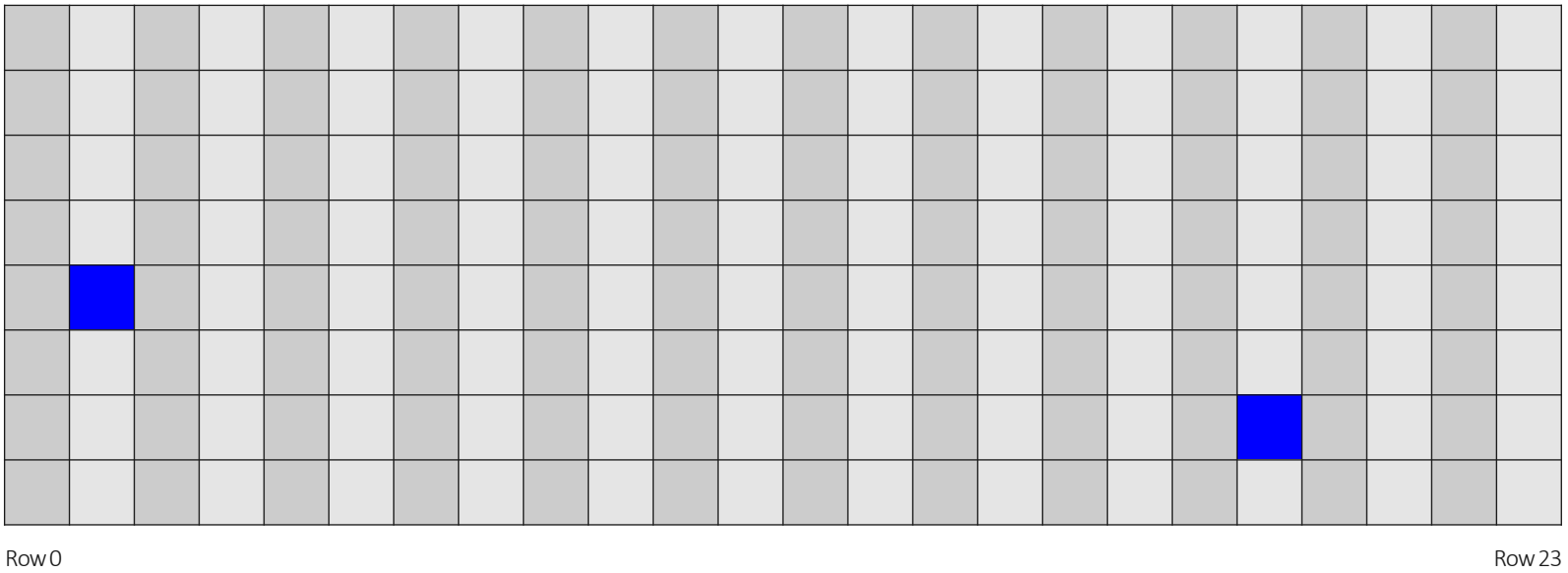
1. Uncached memory access
   → various techniques e.g. `clflush`
2. Fast memory access
   → possible
3. Target specific row
   → reverse-engineer
   mapping functions

Searching for page with flip



Hammering memory locations in different rows

Searching for page with flip



Hammering memory locations in different rows

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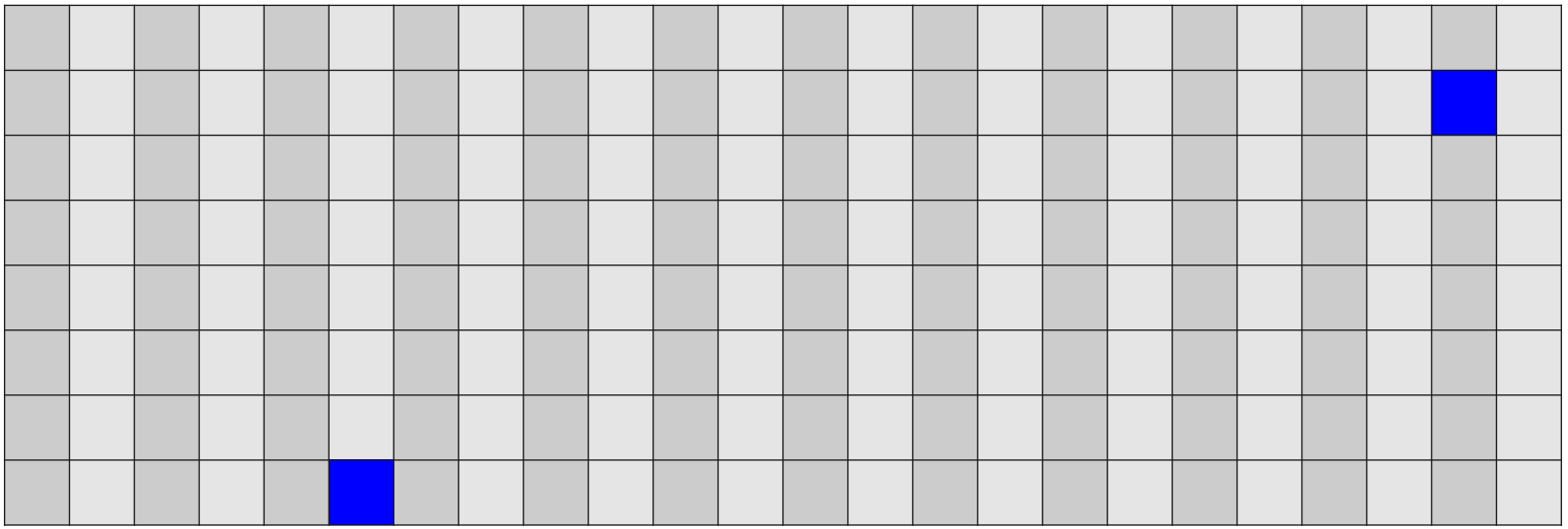
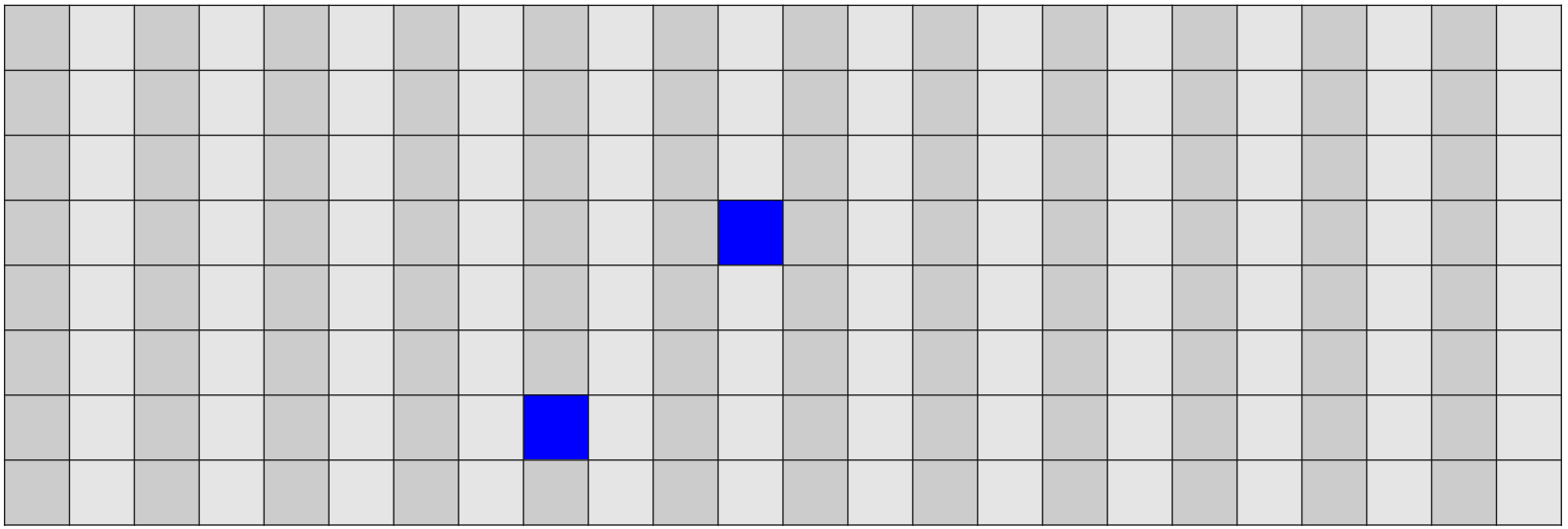


Hammering memory locations in different rows

Searching for page with flip



Hammering memory locations in different rows

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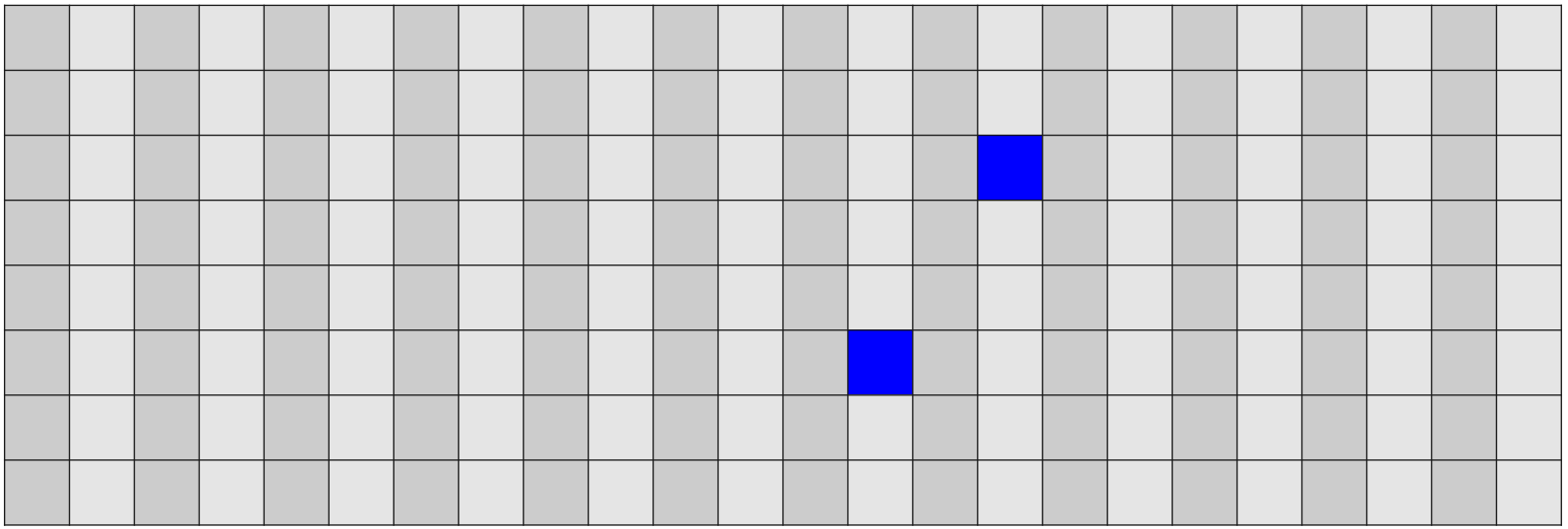
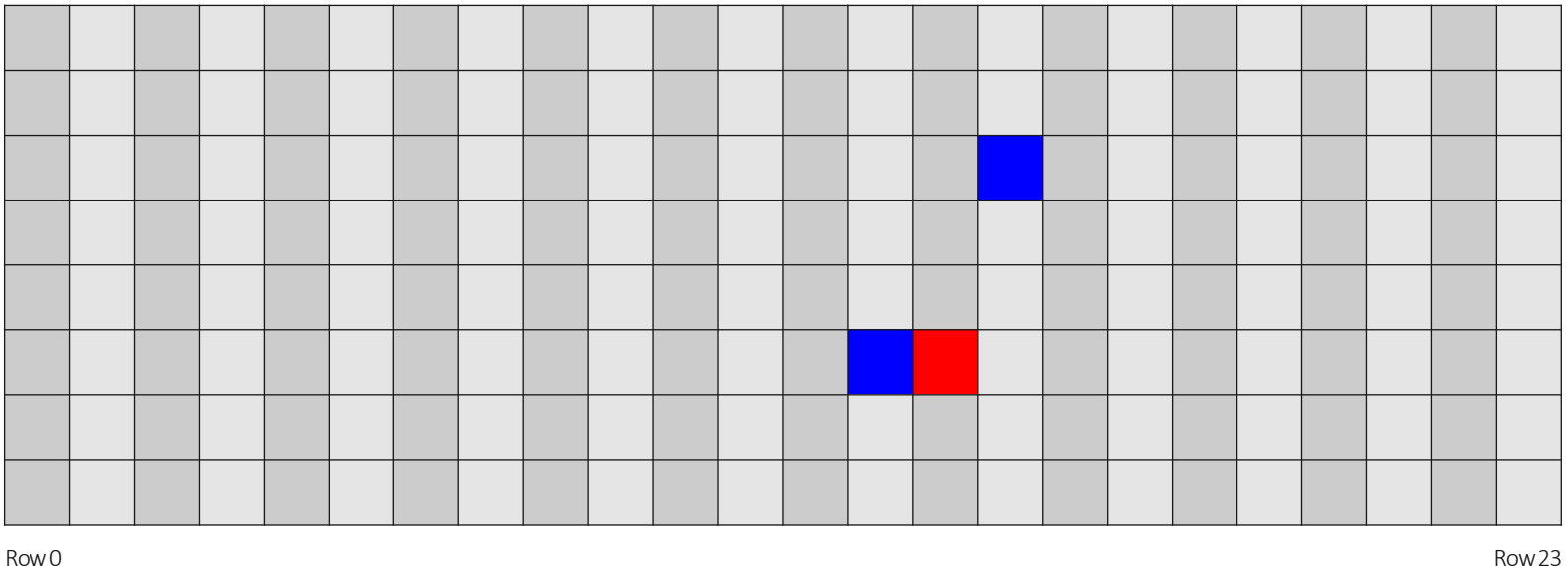


Hammering memory locations in different rows

Searching for page with flip



Hammering memory locations in different rows

- Found a flip

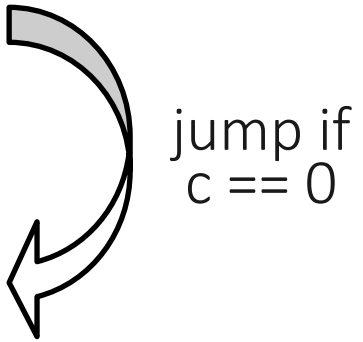
Bypassing PIN check with Rowhammer

```
int c = memcmp(entered_pin, stored_pin, pin_len);

if(c != 0)
{
    return -1;
}

// Code continues ...
```

```
cmp rax, 0
je continue
…
ret
continue:
…
```



Changing code with Rowhammer

Changing code with Rowhammer

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```
cmp rax, 0
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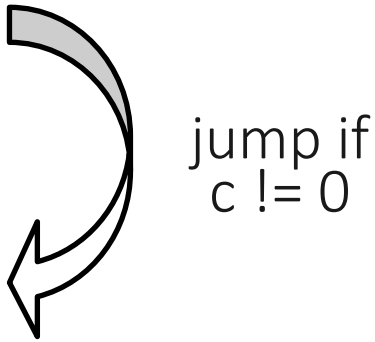
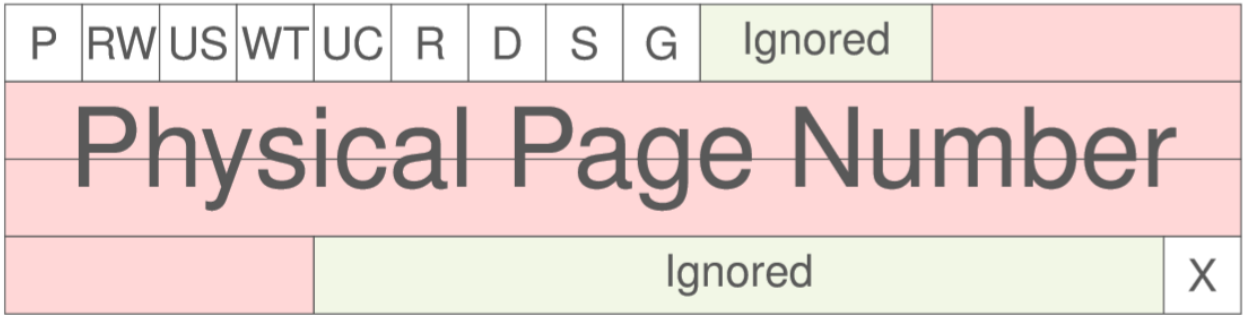
…
ret
continue:
…
```



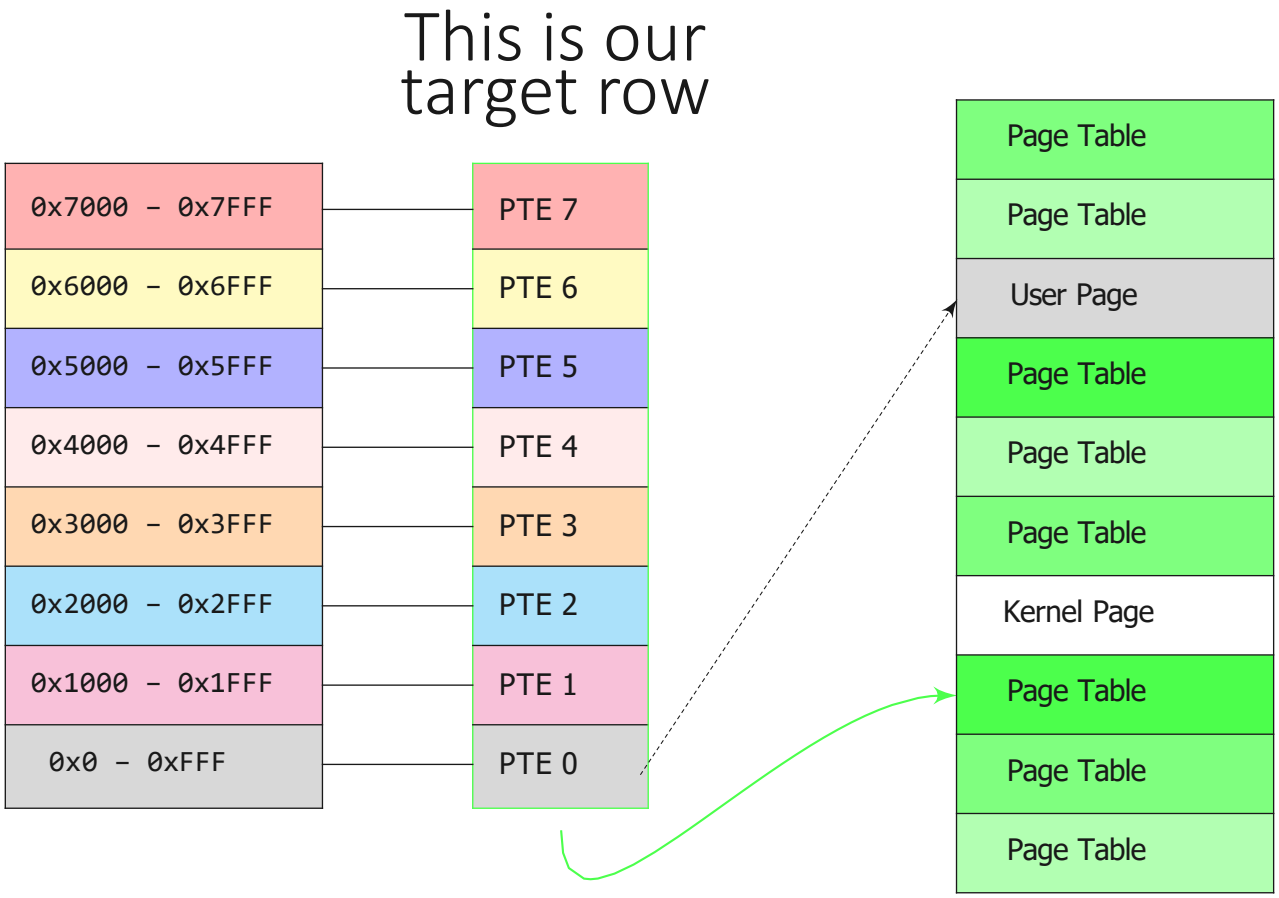
jump if
c != 0

Rowhammer on page tables

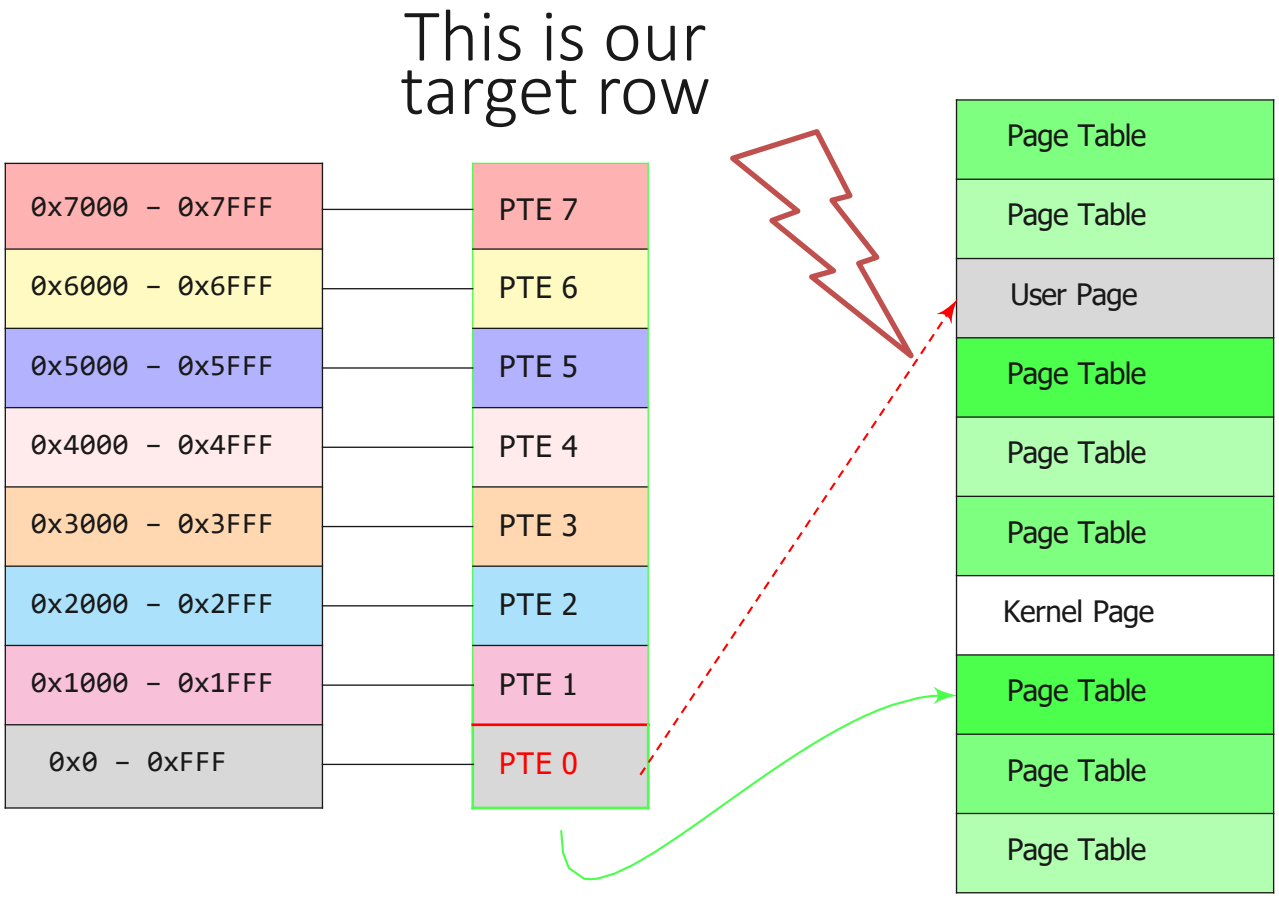
- Find a target location
- Make a page table appear at target location
- Fill physical memory with page tables
- Flip in target PTE likely gives access to a page table → access to physical memory



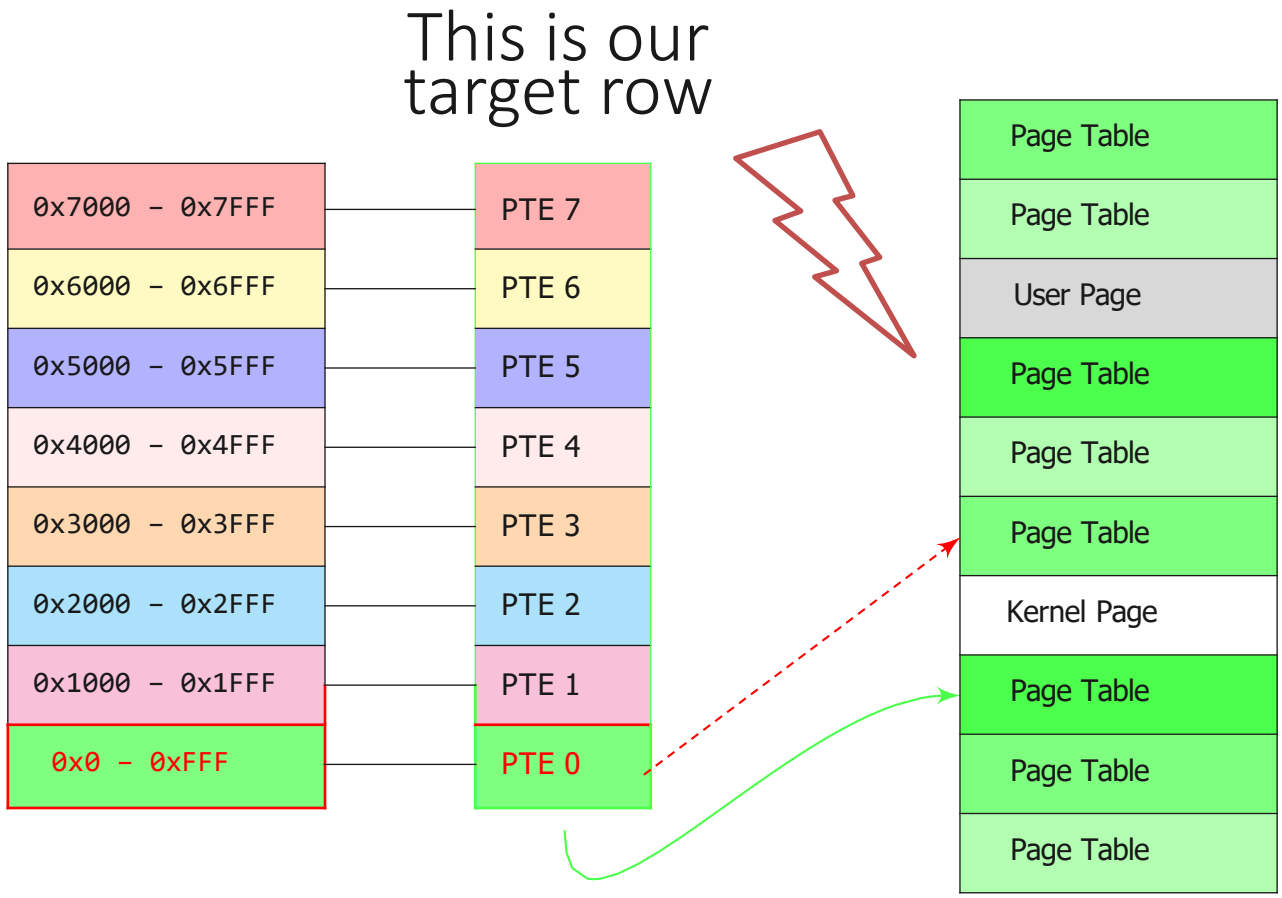
Page table example: before hammering



Page table example: flipping a PTE



Page table example: changing a PTE

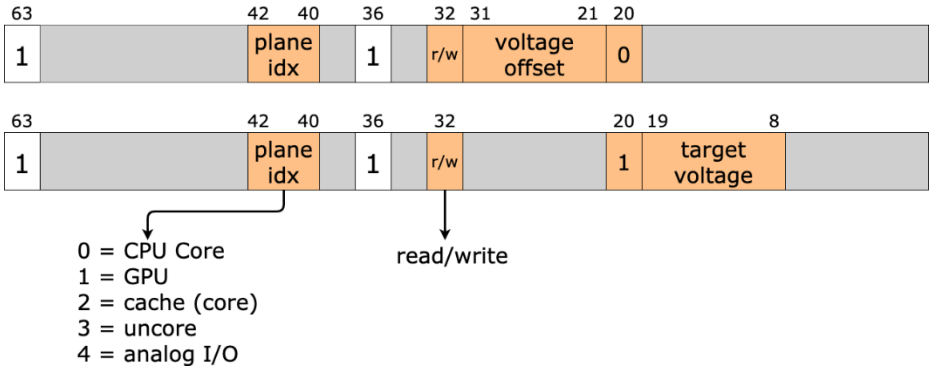


Rowhammer …

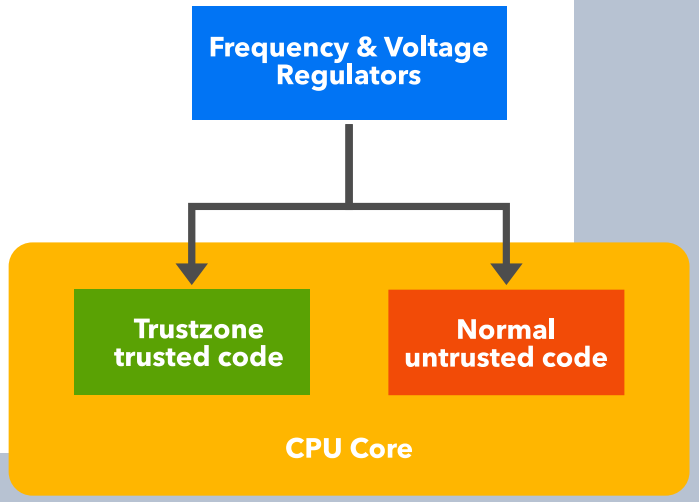
- … can be exploited in various ways
- … works from the browser (Rowhammer.js)

Rowhammer …

- … can be exploited in various ways
- … works from the browser (Rowhammer.js)
- … works on ARM smartphones (Drammer)
- … works for protected DDR4 (TRRespass)
- … can even work on flash memory (SSD):
  https://www.usenix.org/system/files/conference/woot17/woot17-paper-kurmus.pdf
- There are countermeasures, but
  they are not 100% effective



Software-based fault attacks with power management



Power and clock management

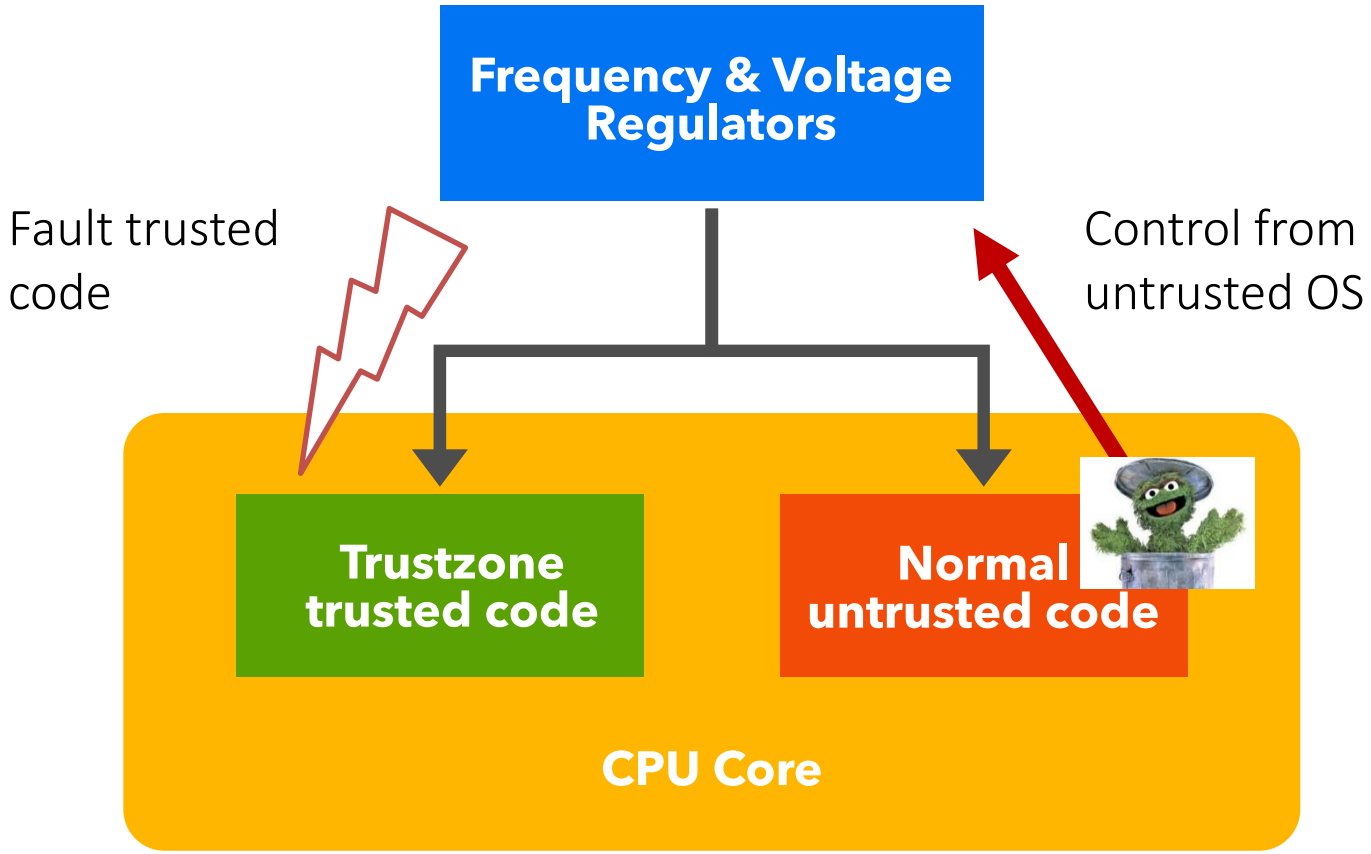
- Modern CPUs continuously adjust their clock frequency and supply voltage to run at minimum power consumption / temperature
- Dynamic Voltage and Frequency Scaling (**DVFS**)
- Too high frequency and/or too low voltage results in faults (typically crashes)
- DVFS is often software-exposed…

CLKscrew and VoltJockey on ARM

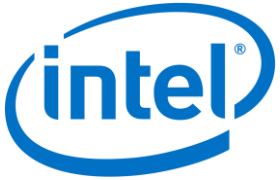
- Smartphones (based on ARM processors) allow clock and voltage control if you're **`root`**
- This can be used to inject controlled faults from privileged software:
  - Tang et al. "CLKSCREW: exposing the perils of security-oblivious energy management", USENIX Security '17
  - Qiu et al. "VoltJockey: Breaching TrustZone by Software-Controlled Voltage Manipulation over Multi-core Frequencies", CCS 2019
- Why do attacks requiring **`root`** matter?

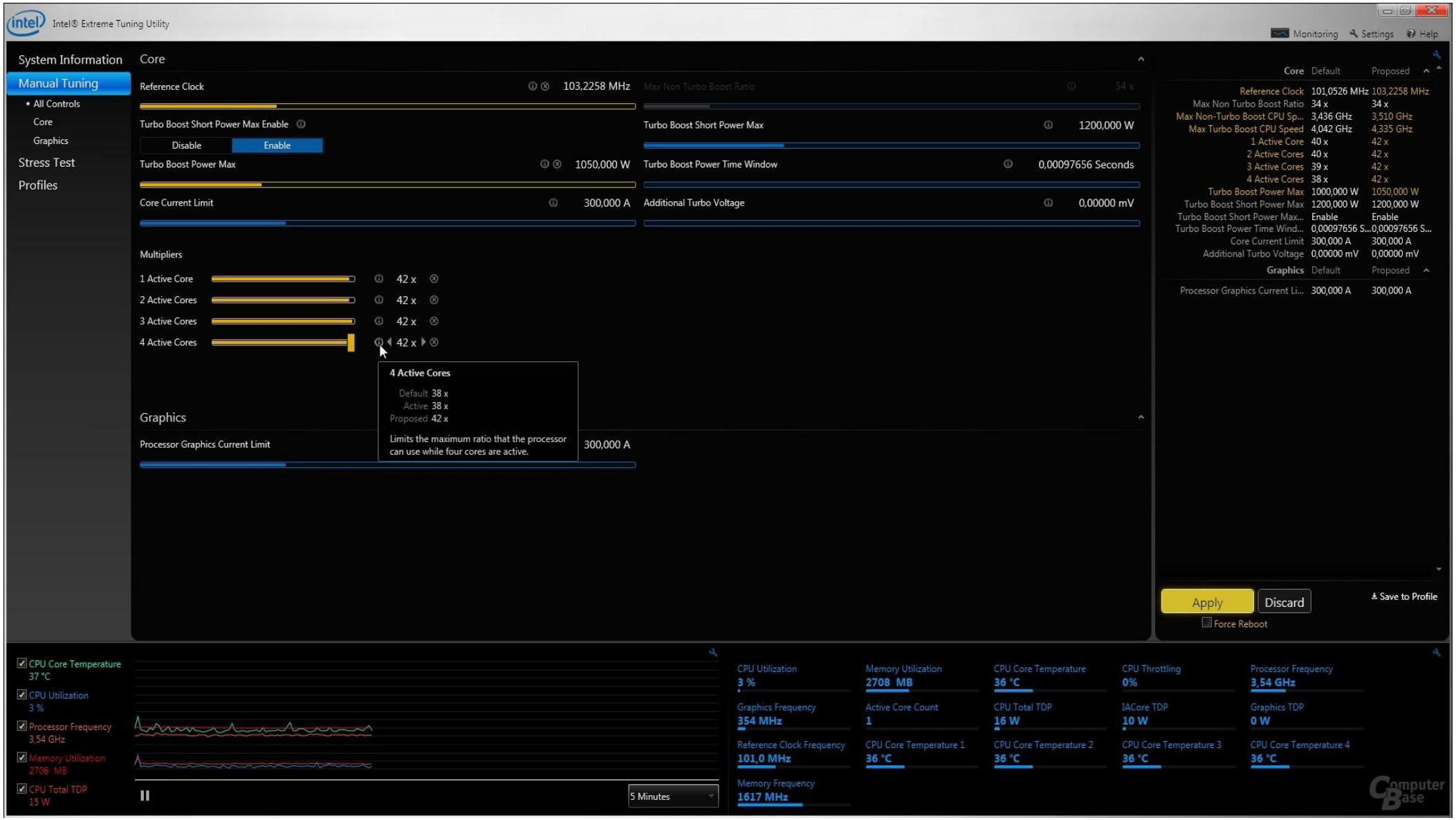
Trusted Execution Environments (TEEs)

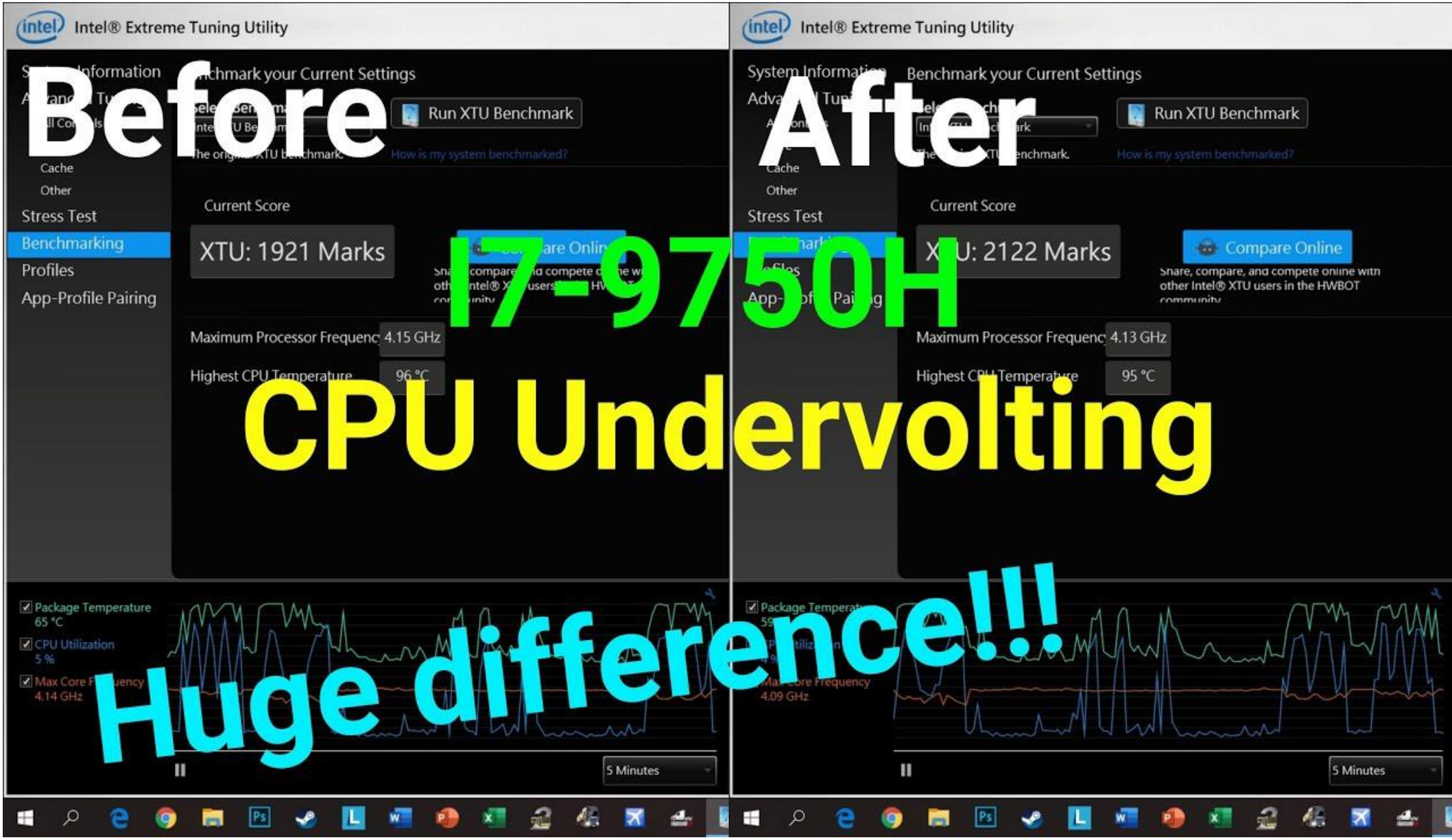
TEEs like ARM Trustzone should protect against attackers with `root` on untrusted OS



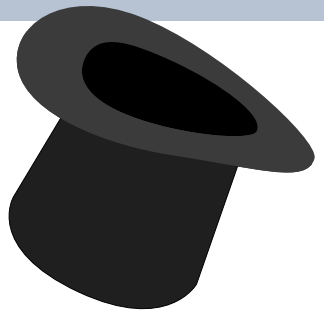
Power management attacks on Trustzone

- Attacker controls untrusted OS …
- … but manages to fault the trusted code in the TEE
- Demonstrated attacks:
  - Break AES (see previous lecture)
  - Load arbitrary code into the Trustzone by faulting a signature check
- What about 

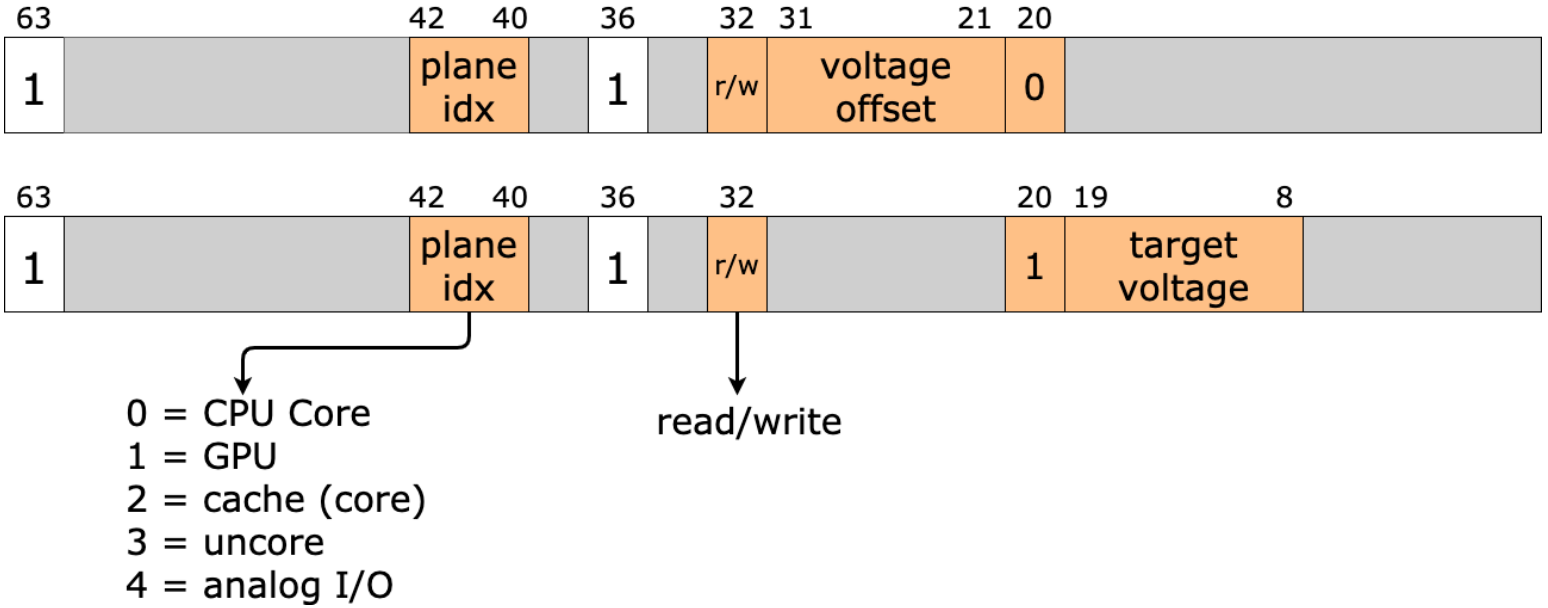

Intel® Extreme Tuning Utility
Monitoring
Settings
Help
System Information
Manual Tuning
• All Controls
Core
Graphics
Stress Test
Profiles
Core
Reference Clock 103,2258 MHz
Max Non Turbo Boost Ratio 34 x
Turbo Boost Short Power Max Enable
Disable
Enable
Turbo Boost Short Power Max 1200,000 W
Turbo Boost Power Max 1050,000 W
Turbo Boost Power Time Window 0,00097656 Seconds
Core Current Limit 300,000 A
Additional Turbo Voltage 0,00000 mV
Multipliers
1 Active Core 42 x
2 Active Cores 42 x
3 Active Cores 42 x
4 Active Cores 42 x
4 Active Cores
Default 38 x
Active 38 x
Proposed 42 x
Limits the maximum ratio that the processor can use while four cores are active.
Graphics
Processor Graphics Current Limit 300,000 A
Core Default Proposed
Reference Clock 101,0526 MHz 103,2258 MHz
Max Non Turbo Boost Ratio 34 x 34 x
Max Non-Turbo Boost CPU Sp... 3,436 GHz 3,510 GHz
Max Turbo Boost CPU Speed 4,042 GHz 4,335 GHz
1 Active Core 40 x 42 x
2 Active Cores 40 x 42 x
3 Active Cores 39 x 42 x
4 Active Cores 38 x 42 x
Turbo Boost Power Max 1000,000 W 1050,000 W
Turbo Boost Short Power Max 1200,000 W 1200,000 W
Turbo Boost Short Power Max... Enable Enable
Turbo Boost Power Time Wind... 0,00097656 S... 0,00097656 S...
Core Current Limit 300,000 A 300,000 A
Additional Turbo Voltage 0,00000 mV 0,00000 mV
Graphics Default Proposed
Processor Graphics Current Li... 300,000 A 300,000 A
Apply
Discard
Save to Profile
Force Reboot
CPU Core Temperature 37 °C
CPU Utilization 3 %
Processor Frequency 3,54 GHz
Memory Utilization 2708 MB
CPU Total TDP 15 W
5 Minutes
CPU Utilization 3 %
Memory Utilization 2708 MB
CPU Core Temperature 36 °C
CPU Throttling 0%
Processor Frequency 3,54 GHz
Graphics Frequency 354 MHz
Active Core Count 1
CPU Total TDP 16 W
IACore TDP 10 W
Graphics TDP 0 W
Reference Clock Frequency 101,0 MHz
CPU Core Temperature 1 36 °C
CPU Core Temperature 2 36 °C
CPU Core Temperature 3 36 °C
CPU Core Temperature 4 36 °C
Memory Frequency 1617 MHz
Computer Base


Before
After
i7-9750H
CPU Undervolting
Huge difference!!!
Intel® Extreme Tuning Utility
Benchmark your Current Settings
Run XTU Benchmark
How is my system benchmarked?
Cache
Other
Stress Test
Benchmarking
Profiles
App-Profile Pairing
Current Score
XTU: 1921 Marks
Maximum Processor Frequenc 4.15 GHz
Highest CPU Temperature 96 °C
Compare Online
Package Temperature
65 °C
CPU Utilization
5 %
4.14 GHz
5 Minutes
XTU: 2122 Marks
Maximum Processor Frequenc 4.13 GHz
Highest CPU Temperature 95 °C
Share, compare, and compete online with other Intel® XTU users in the HWBOT
4.09 GHz

How does that work?



`msr 0x150`

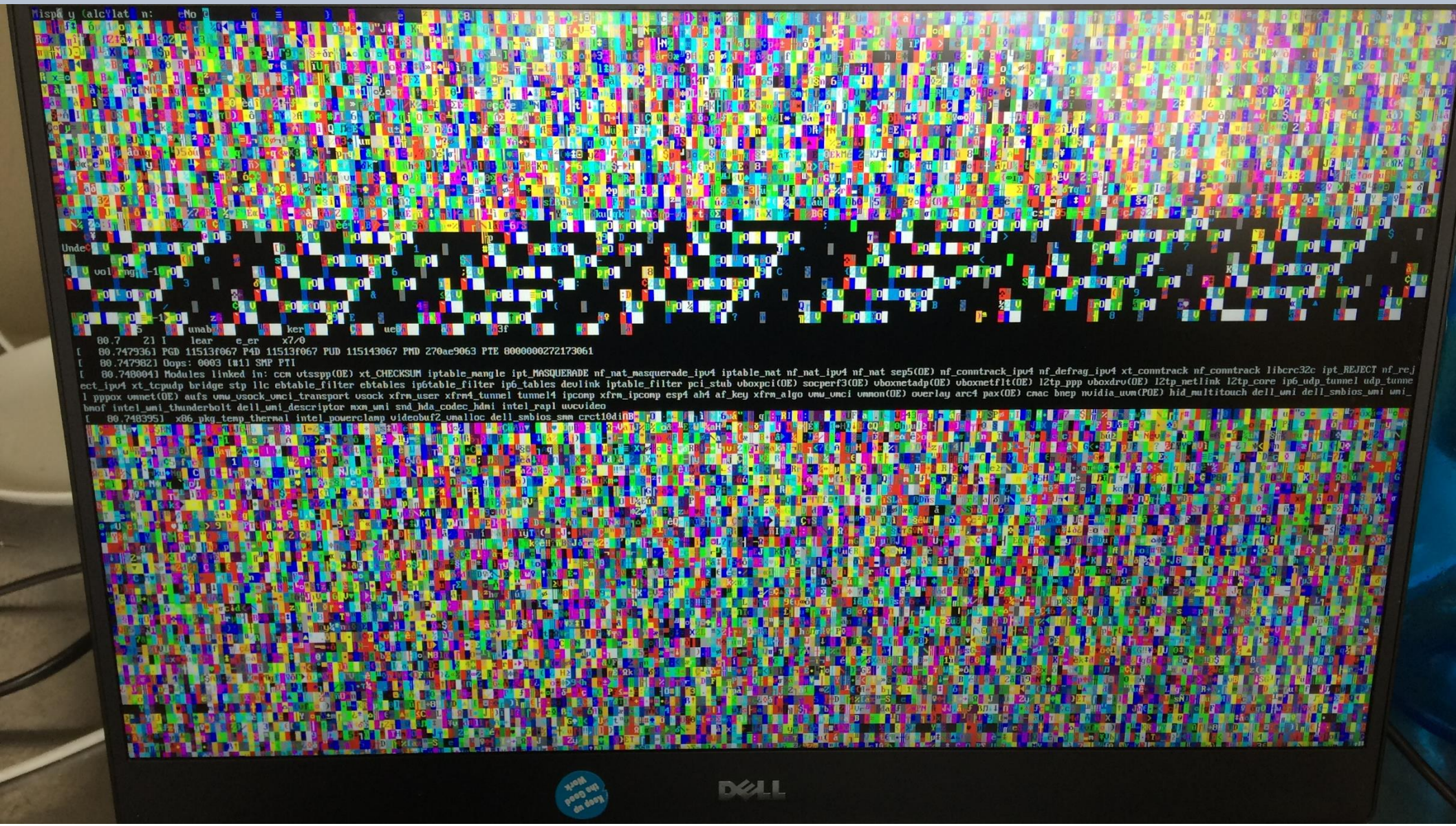


A simple proof-of-concept

```
uint64_t multiplier = 0x1122334455667788;
uint64_t correct    = 0xdeadbeef * multiplier;
uint64_t var        = 0xdeadbeef * multiplier;

// start undervolting
while ( var == correct )
{
    var = 0xdeadbeef * multiplier;
}
// stop undervolting

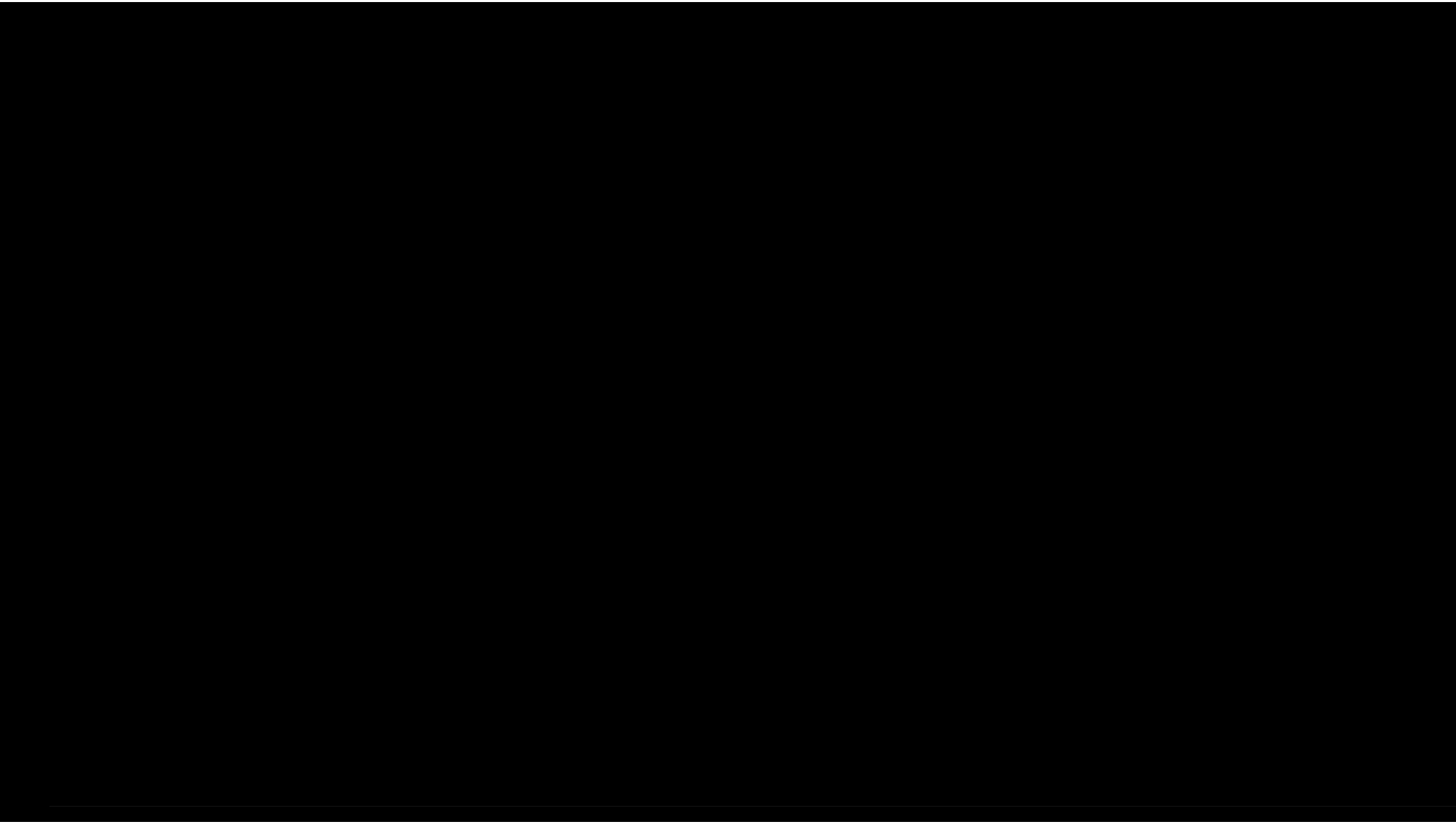
// Can we ever get here? Yes!!
uint64_t flipped_bits = var ^ correct;
```


[80.7479361 PGD 11513f067 P4D 11513f067 PUD 115143067 PMD 270ae9063 PTE 8000000272173061
[80.7479821 Oops: 0003 [#1] SMP PTI
[80.7480041 Modules linked in: ccm vtsspp(OE) xt_CHECKSUM iptable_mangle ipt_MASQUERADE nf_nat_masquerade_ipv4 iptable_nat nf_nat_ipv4 nf_nat sep5(OE) nf_conntrack_ipv4 nf_defrag_ipv4 xt_conntrack nf_conntrack libcrc32c ipt_REJECT nf_reject_ipv4 xt_tcpudp bridge stp llc ebtable_filter ebtables ip6table_filter ip6_tables devlink iptable_filter pci_stub vboxpci(OE) socperf3(OE) vboxnetadp(OE) vboxnetflt(OE) l2tp_ppp vboxdrv(OE) l2tp_netlink l2tp_core ip6_udp_tunnel udp_tunnel pppox vmnet(OE) aufs vmw_vsock_vmci_transport vsock xfrm_user xfrm4_tunnel tunnel4 ipcomp xfrm_ipcomp esp4 ah4 af_key xfrm_algo vmw_vmci vmmon(OE) overlay arc4 pax(OE) cmac bnep nvidia_uvm(POE) hid_multitouch dell_wmi dell_smbios_wmi wmi_bmof intel_wmi_thunderbolt dell_wmi_descriptor mxm_wmi snd_hda_codec_hdmi intel_rapl uvcvideo
[80.7483951 x86_pkg_temp_thermal intel_powerclamp videobuf2_vmalloc dell_smbios_smm crct10dif
DELL

Live-Demo

"Anything that can go wrong, will go wrong"

Plundervolt: Multiplications



Carmen Crincoli, but Fhqwhgads
@CarmenCrincoli

"Plundervolt" sure does have a catchy name and
an exploit that
running

1:59 AM · D

Larry Osterman
@osterman

Replying to @CarmenCrincoli and @MT6572A

Since the threat model for SGX assumes that the attacker has root access (and I believe also has physical control over the hardware), this is actually a bigger deal than you make it out to be.

2:07 AM · Dec 11, 2019 · Twitter Web App

Some attacks

- Key extraction attack on RSA and AES-NI

Faulting CRT-RSA

```
// Start undervolting
uint8_t rsa_dec_ecall(int iterations)
{
    //Waitforfirstfault
    trigger_fault(iterations);

    //Actualdecryption
    ippsRSA_Decrypt(ct,dec,pPrv,scratchBuffer);
}
// Stop undervolting
```

Faulting CRT-RSA

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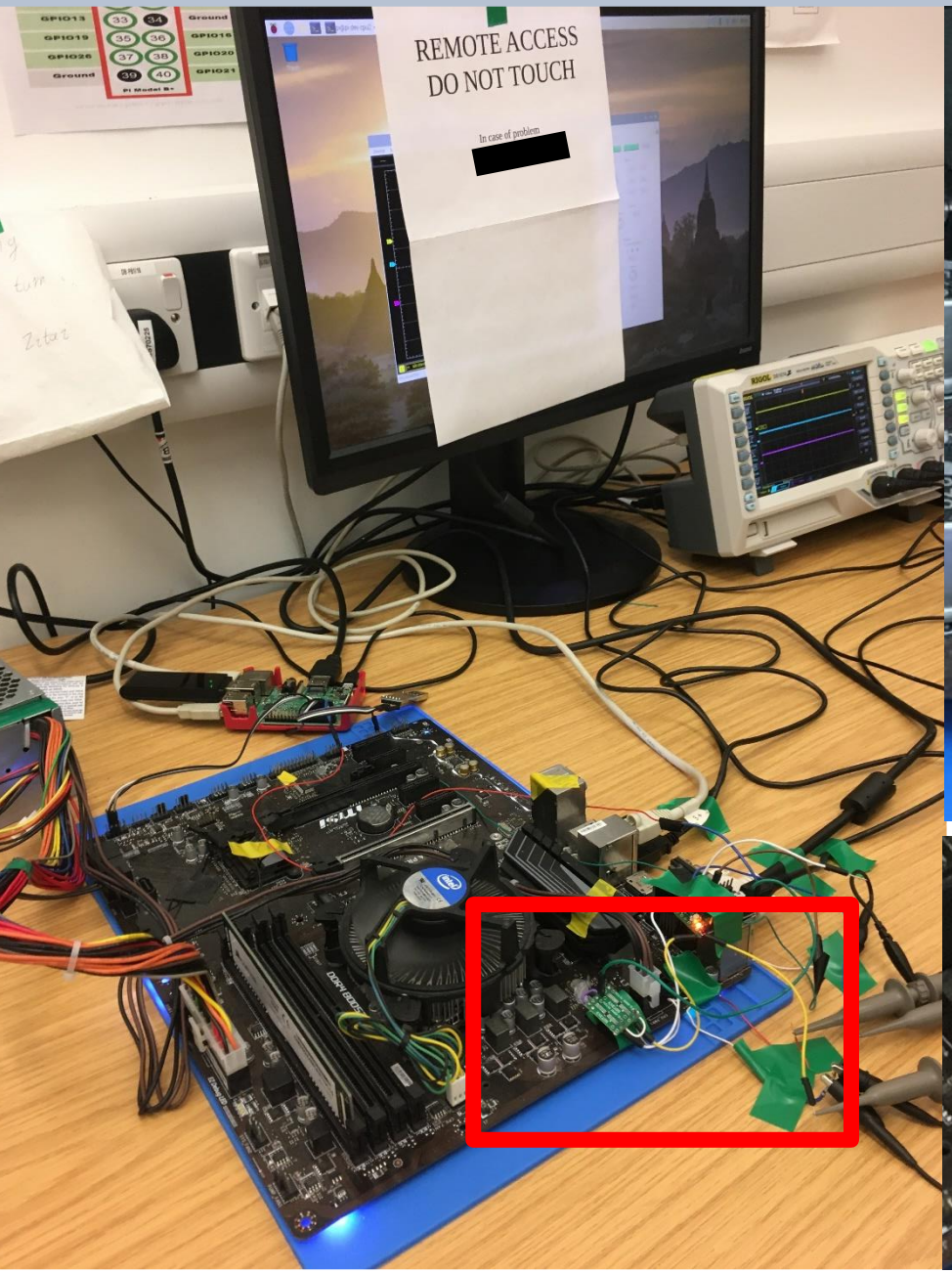
```

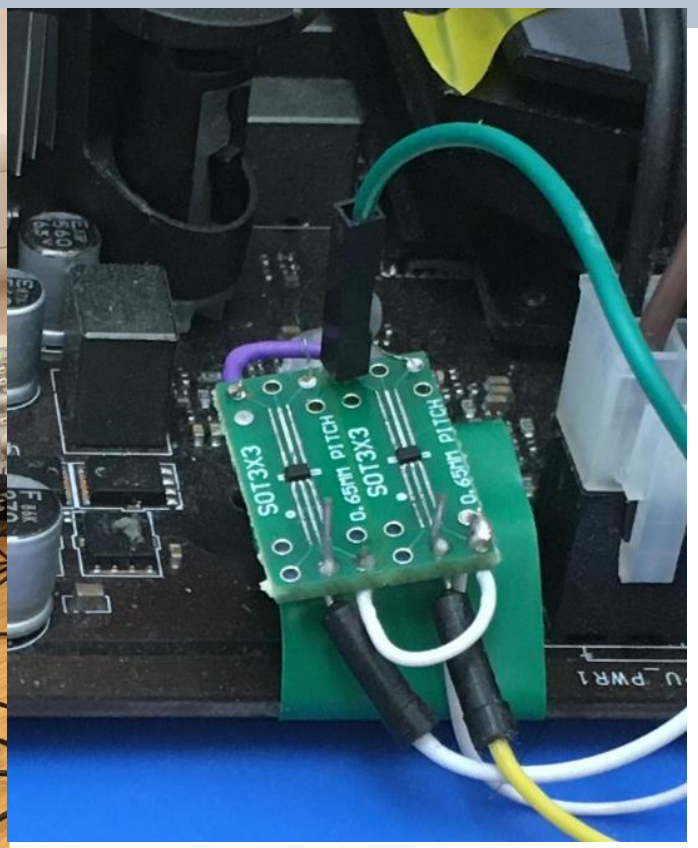
- Bellcore attack: gcd($M' - M$, n)
- Lenstra attack: gcd($(M')^e - C$, n)
- yields p or q (and dividing n by it gives the other)

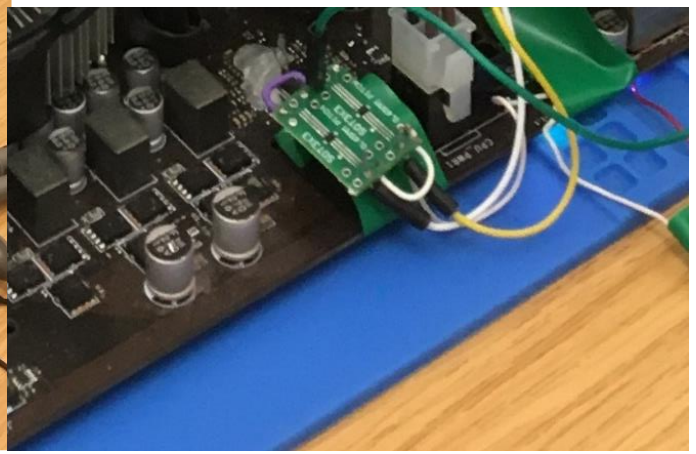
Some attacks

- Key extraction attack on RSA and AES-NI
- Induce memory safety vulnerabilities in bug-free enclave code
- Responsible disclosure: Intel released microcode update with fix for CVE-2019-11157 in Dec 2019
- Fix: turn off software undervolting interface in MSR 0x150
  (what about hardware attacks?)

Hardware-level undervolting against SGX







Take-home points

- Fault injection is not limited to hardware attackers: faults can be injected remotely
- Fault injection is not limited to crypto code: generic code can be broken (e.g. regarding memory safety)
- Countermeasures against this are even harder: PCs are general purpose systems, not high-security smartcards
- Fault injection is there to stay

Thanks!

Questions / discussion?

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